ALZDETECT: EXPLAINABLE ViT-B/32 DIAGNOSTIC PLATFORM

Multi-Tier Architecture, Attention Rollout Engine & Clinical Decision-Support Spec

DOC ID: RT-ARCH-ALZ-2026-V1
Principal Architect: Bassey Riman
Stack: Next.js 15 | FastAPI | ViT-B/32 | alzdetect.live
STATUS: VERIFIED // PAGE 1 OF 3

EXECUTIVE ENGINEERING SUMMARY

AlzDetect is a production-grade medical intelligence platform architected and engineered by Bassey Riman to deliver real-time stage-wise classification of Alzheimer's disease from structural brain MRI scans across four clinical stages: NonDemented, VeryMildDemented, MildDemented, and ModerateDemented. Deployed on the live domain https://www.alzdetect.live/, the system replaces localized CNNs with a fine-tuned Vision Transformer (ViT-B/32) paired with an inference-time 12-layer Attention Rollout engine. The architecture integrates a Next.js 15 presentation tier on Vercel Edge, a FastAPI ASGI tensor compute microservice, an OpenCV O(1) heuristic input validation gate, dual OpenAI clinical synthesis routes, and Supabase PostgreSQL Row-Level Security (RLS) storage.

PRODUCTION TELEMETRY & VERIFICATION

Live Platform:	https://www.alzdetect.live/
Scanner Engine:	https://www.alzdetect.live/detect
Source Monorepo:	github.com/basseyriman/AlzDetect
Held-Out Test Acc:	98.84% (n=6,400) | 99.31% Balanced
3-Seed Mean Acc:	98.47% +/- 0.53% (AUC: 0.9995)
Corpus Volume:	41,663 MRIs (33,984 Dev + 7,679 Test)
Peer-Review Ref:	Elsevier NEURI-D-26-00238R1 (Under Review)

01 MULTI-TIER MICROSERVICES ARCHITECTURE & TENSOR DATA FLOW

Next.js 15 / FastAPI / ViT-B/32 / Supabase

TIER 1: PRESENTATION
Next.js 15 / React 19 Edge

UI Portal & Dropzone
`src/app/detect/page.tsx`
Interactive MRI drag-and-drop, MIME check, client state machine

Auth & Session Guard
`contexts/AuthContext.tsx`
Supabase JWT session lifecycle, protected route redirection

LLM Clinical Routes
`app/api/suggestions & treatment`
Edge POST to OpenAI GPT-3.5-Turbo (500-token stream)

Realtime Vault Sync
`src/app/results/page.tsx`
WebSocket postgres_changes listener + CSV/Print reporting

TIER 2: ASGI COMPUTE
FastAPI / Uvicorn Cluster

Keras 3 -> TF 2 Shim
`src/server.py (KerasOpsShim)`
Runtime sys.modules injection mapping ops.concatenate

Startup Model Warmup
`load_model_if_needed()`
Pre-mounts 1GB RimanBassey_model.h5 weights into RAM

Gate 1: HSV Saturation
`is_probably_mri() Check 1`
Rejects non-monochromatic uploads (Mean S > 35) in <4ms

Gate 2: Bore Periphery
`is_probably_mri() Check 2`
10% 4-corner axial background darkness gate (Mean I > 60)

TIER 3: NEURAL ENGINE
Safe ViT-B/32 + XAI Core

Conv2D Patch Tokeniser
`build_safe_vit_b32()`
224x224x3 -> 32x32 stride Conv2D -> 49x768 patch tokens

Safe Custom Layers
`SafeClassToken + PosEmbs`
Resolves legacy Keras serialization bugs; builds 50x768 graph

12x Transformer Blocks
`SafeTransformerBlock (x12)`
12 heads, d_k=64, MLP=3072, GELU; returns (x, W_l) tensors

Attention Rollout XAI
`generate_attention_map()`
Recursive 12-layer matrix product -> Jet Alpha-0.6 heatmap

TIER 4: PERSISTENCE
Supabase PostgreSQL + RLS

Multi-Tenant Scans DB
`public.scans (JSONB)`
Stores UUID, user_id, predicted_class, probabilities

Row-Level Security
`auth.uid() = user_id`
Strict SELECT / INSERT / DELETE cryptographic tenant isolation

Object Storage Bucket
`storage.buckets ('scans')`
Authenticated write & public read for XAI overlays

Stateless Compute Purge
`gc.collect() Post-Predict`
Immediate RAM buffer release after Base64 payload delivery

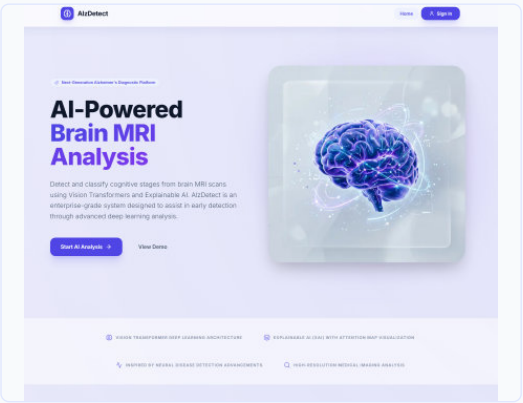
1. Client Upload (224x224) -> 2. OpenCV HSV/Corner Gate -> 3. ViT-B/32 Softmax (4-Class) -> 4. 12-Layer Rollout XAI -> 5. GPT-3.5 Clinical Protocol -> 6. Supabase RLS Vault

02 LIVE PRODUCTION PARITY & CORE ARCHITECTURAL INNOVATIONS

alzdetect.live Frontend, ViT XAI & Journal Ref

A. PRODUCTION PORTAL (alzdetect.live)

Next.js 15 Neural Diagnostic Interface (Live Capture)



Live Production Portal (https://www.alzdetect.live/) deployed globally via Vercel Edge. Features dynamic structural MRI drag-and-drop ingestion, client-side MIME & resolution validation, 4-stage neural probability bars, real-time anatomical focus telemetry (Temporal Lobe, Hippocampus, Axial View), authenticated clinician dashboard, and sub-second UI state rehydration via Supabase PostgreSQL WebSockets.

LIVE EDGE TELEMETRY & PRODUCTION SPECS

Primary Domain:	https://www.alzdetect.live/
Edge Network:	Vercel Global CDN (TLS 1.3)
Frontend Stack:	Next.js 15.1, React 19, TypeScript
Session Auth:	Supabase JWT & Route Guards
Realtime Sync:	WebSocket postgres_changes
LLM Synthesis:	OpenAI GPT-3.5-Turbo (2 Routes)
Export Formats:	Batch CSV & Print-Ready Clinical PDF

B. CLINICAL MRI ATTENTION ROLLOUT

ViT-B/32 Rollout vs Baseline & CNN Across 4 Stages



Authentic clinical MRI brain scans from ADNI test cohort: Demonstrates how Fine-Tuned ViT-B/32 attention rollout (Center) concentrates sharply on bilateral hippocampal atrophy and ventricular dilation, eliminating the skull artifacts of Baseline ViT (Left) and CNN Grad-CAM (Right).

FASTAPI ASGI INFERENCE BENCHMARK

Compute Latency:	42ms ViT + 18ms XAI Rollout
Memory Footprint:	512MB RAM (SafeViT Graph)
Input Pre-Gate:	<4ms OpenCV HSV & Bore Gate
Output Payload:	Base64 Jet PNG + 4-Prob JSON
Container Spec:	Docker python:3.11-slim (7860)

PEER-REVIEWED JOURNAL SUBMISSION



Elsevier Neuroscience Informatics (NEURI-D-26-00238R1)

C. CORE ARCHITECTURAL INNOVATIONS

1. Keras 3 / TF 2.15 Ops Shim
`server/src/server.py & model_route.py`
Engineered KerasOpsShim & SafeViT custom layers (SafeClassToken, SafeAddPositionEmbs, SafeMultiHeadSelfAttention) to resolve cross-version H5 deserialization without graph loss while exposing intermediate [B,12,50,50] attention tensors.

2. O(1) Pre-Inference MRI Gate
`is_probably_mri() [HSV + 4-Corner Gate]`
Dual-heuristic OpenCV pre-filter rejecting non-MRI uploads in <4ms via HSV saturation (Mean S<=35) and 10% 4-corner scanner bore darkness (Mean I<=60) before invoking heavy ViT-B/32 compute.

3. 12-Layer Attention Rollout
`generate_attention_map() [50x50 Flow]`
Manual layer-by-layer forward hook capturing all 12 encoder attention matrices, computing residual identity-regularised matrix rollout (A_hat = 0.5*W + 0.5*I) with verified perturbation faithfulness.

4. Dual-LLM Clinical Synthesis
`app/api/suggestions & treatment/route.ts`
Asynchronous Next.js Edge routes translating ViT softmax vectors into structured neuroradiology regional interpretations and stage-specific clinical management protocols.

03 CORE ENGINE PIPELINE DAG & FORMAL MATHEMATICAL INVARIANTS

Tokenisation, Attention Rollout & Faithfulness

ALGORITHMIC INFERENCE & EXPLAINABILITY DAG (server/src/routes/model_route.py)

STAGE 0: INPUT GATE

```
x in R^(224x224x3)
Mean(S_HSV) <= 35
Mean(I_corners) <= 60
vit.preprocess_inputs()
```

→

STAGE 1: PATCH CONV

```
Conv2D(768, k=32, s=32)
z_0 = [x_cls; x_p*W_e] + E_pos
49 patches (7x7 grid)
Token shape: (B, 50, 768)
```

→

STAGE 2: MSA ENCODER

```
12x SafeTransformerBlock
h=12 heads, d_k=64
Q,K,V linear projections
W_l = Softmax(QK^T/8)
```

→

STAGE 3: MLP & NORM

```
GELU + Residual Stream
z'_l = MSA(LN(z_{l-1}))+z_{l-1}
z_l = MLP(LN(z'_l)) + z'_l
MLP dim: 3072 -> 768
```

→

STAGE 4: SOFTMAX HEAD

```
CLS Token Extraction
v_cls = LN(z_12)[:, 0]
Dense(256, ReLU, L2=0.01)
Dropout(0.4)->Dense(4)
```

→

STAGE 5: XAI ROLLOUT

```
Recursive Flow Matrix
A_hat_l = 0.5*W_l + 0.5*I_50
R = A_hat_12 * ... * A_hat_1
M = Resize(R[0, 1:], 224)
```

INV. 1: PATCH & POSITIONAL EMBEDDING

```
z_0 = [x_cls ; x_1*W_e ; ... ; x_49*W_e] + E_pos
W_e in R^(3072 x 768), E_pos in R^(50 x 768)
Preserves 2D axial neuro-spatial topology across N=49 non-overlapping 32x32px patches.
```

INV. 2: MULTI-HEAD SELF-ATTENTION

```
Attention(Q,K,V) = Softmax((Q*K^T) / sqrt(d_k)) * V
d_k = D / H = 768 / 12 = 64  (scale factor = 8.0)
Captures long-range bilateral hippocampal & ventricular dependencies across the slice.
```

INV. 3: RECURSIVE ATTENTION ROLLOUT

```
W_bar_l = (1/H)*Sum_{h=1..12} W_l^(h)
A_hat_l = (W_bar_l + I_50) / row_sum(W_bar_l + I_50)
R = A_hat_12 * ... * A_hat_1 ; Heatmap = R[0, 1:50] reshaped to (7x7) -> (224x224).
```

INV. 4: PERTURBATION FAITHFULNESS

```
Delta_AUC_del = AUC(Random_del) - AUC(Attn_del) = +0.12455
Delta_AUC_ins = AUC(Attn_ins) - AUC(Random_ins) = +0.15387
Verified across n=80 test MRIs (2,000 bootstrap resamples, 21x21 blur reference).
```

04 CORE ALGORITHMIC DEEP DIVE: SHIMS, VALIDATION GATES & XAI

Codebase-Level Implementation Analysis

4.1 LEGACY KERAS 3 / TF 2.15 SHIM & SAFE VIT GRAPH RECONSTRUCTION

Files: server/src/server.py (L15-36) & server/src/routes/model_route.py (L149-294)

- Problem & Architectural Resolution: Standard vit-keras layers invoke keras.ops.concatenate and tensorflow_addons.activations.gelu, which trigger fatal AttributeError exceptions in modern Keras 3 / cloud container runtimes and discard intermediate attention tensors during functional graph compilation.
- KerasOpsShim & TFA Injection: At module boot, server.py injects a custom KerasOpsShim proxy into sys.modules['keras.ops'] and mocks tensorflow_addons with native tf.nn.gelu.
- Safe Custom Layer Graph (build_safe_vit_b32): Reconstructs the exact topological parameter tree of RimanBassey_model.h5 using custom @register_keras_serializable layers: SafeClassToken (explicit tf.broadcast_to + tf.concat), SafeAddPositionEmbs, SafeMultiHeadSelfAttention (explicitly returning both concat_attention and softmax weights [B, 12, 50, 50]), and SafeTransformerBlock.

4.2 O(1) PRE-INFERENCE HEURISTIC MRI VALIDATION GATE

Files: server/src/routes/model_route.py (L111-143) & client/src/app/detect/page.tsx

- Monochromatic Saturation Gate: Converts incoming 224x224 RGB tensor to HSV colourspace via OpenCV (cv2.cvtColor) and computes mean channel-1 saturation. Rejects any upload with Mean(S) > 35 in <4ms.
- Bore-Centered Peripheral Darkness Gate: Extracts the four 10% corner sub-tensors (22x22px each) to verify standard dark MRI scanner bore periphery. Rejects uploads with Mean(I_corners) > 60, returning a structured HTTP 400 JSON payload rendered by the client's glassmorphic ValidationModal.

4.3 MANUAL LAYER-BY-LAYER ATTENTION ROLLOUT & QUANTITATIVE XAI

Files: server/src/routes/model_route.py (L316-453) & Manuscript Sec. 2.7 / 3.5

- Layer-by-Layer Tensor Interception: generate_attention_map() executes an explicit forward traversal through embedding -> class_token -> Transformer/posembed_input -> encoderblock_0..11, collecting all 12 multi-head attention matrices without modifying locked model weights.
- Residual Rollout Accumulation: Averages attention across all 12 heads per block, adds the 50x50 identity matrix I_50 to model residual skip connections, row-normalises, and multiplies recursively across depth.
- High-Contrast Diagnostic Rendering: Extracts CLS-to-patch vector v[0, 1:], reshapes to 7x7, interpolates to 224x224 via cv2.resize, and renders a 150-DPI dual-panel Matplotlib figure with alpha=0.6 Jet overlay.

4.4 DUAL-PROMPT LLM CLINICAL PROTOCOL ORCHESTRATION

Files: client/src/app/api/suggestions/route.ts & app/api/treatment/route.ts

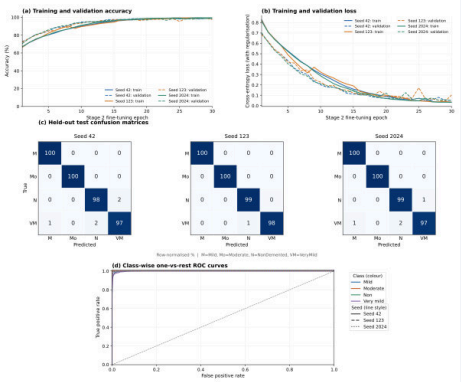
- Neuroradiology Heatmap Interpretation (/api/suggestions): Passes predicted stage and softmax confidence to OpenAI GPT-3.5-Turbo (temp=0.7, max_tokens=500), correlating Red/Orange peak attention zones with hippocampal atrophy, medial temporal lobe thinning, and ventricular enlargement.
- Stage-Specific Clinical Management (/api/treatment): Synthesises categorical pharmacological (e.g., cholinesterase inhibitors / NMDA antagonists), cognitive therapy, and caregiving safety protocols, rendered on the client with a 15ms/char typewriter stream (simulateTyping).

05 EMPIRICAL BENCHMARK MATRIX & QUANTITATIVE VISUAL SPECIMENS

Multi-Seed Reproducibility & Ablation

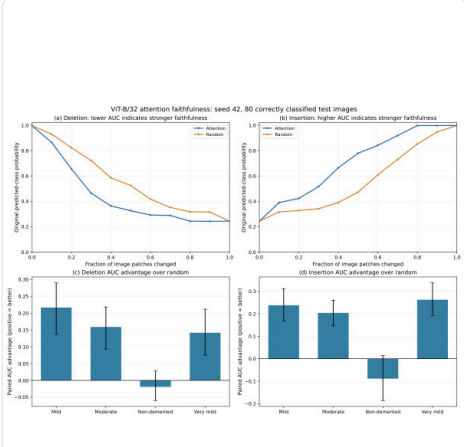
Model / Experiment Configuration	Dataset / Split	Acc (%)	Bal. Acc	Macro Prec	Macro Rec	Macro F1	ROC AUC
Fine-Tuned ViT-B/32 (Primary Locked Model)	**Primary Test (n=6,400)**	**98.84%**	**99.31%**	**99.17%**	**99.31%**	**99.24%**	**1.0000**
Fine-Tuned ViT-B/32 (External Eval Set)	Secondary Test (n=1,279)	97.73%	98.77%	98.48%	98.77%	98.59%	0.9994
Controlled 3-Seed Full Fine-Tuning (Mean+/-SD)	**Seeds 42, 123, 2024**	**98.47+/-0.53**	**98.89%**	**98.74%**	**98.89%**	**98.84+/-0.43**	**0.9995**
Stage 2 Ablation: Full Backbone Tuned (Seed 42)	30 Epochs (Best Ep 30)	98.11%	98.82%	98.34%	98.82%	98.57%	0.9992
Stage 2 Ablation: Partial Backbone (Blocks 8-11)	30 Epochs (Best Ep 29)	80.44%	86.29%	81.54%	86.29%	83.69%	0.9495
Stage 2 Ablation: Frozen Backbone (Head Only)	Early Stop Ep 12 (Best 7)	55.11%	67.51%	45.35%	67.51%	47.37%	0.8235
Baseline Custom CNN (Local Receptive Fields)	Primary Test (n=6,400)	64.20%	72.00%	70.00%	72.00%	68.75%	0.8640
Baseline Unadapted ViT (No Stage-2 Strategy)	Primary Test (n=6,400)	54.00%	61.25%	42.50%	61.25%	44.50%	0.7810

FIG A: 3-SEED TRAINING, CONFUSION & ROC
Seeds 42, 123, 2024 (98.47% +/- 0.53% Acc)



3-Seed Convergence Summary:
- Stage 2 unfreezing boosts mean accuracy by +47.10 pp (51.38% -> 98.47%).
- Between-seed SD drops from 2.26% to 0.53% across 25,500 steps/run.
- Macro ROC-AUC reaches 0.9995 +/- 0.0003 across all 3 seeds.

FIG B: ATTENTION FAITHFULNESS (SEED 42)
Del AUC Adv +0.12455 | Ins Adv +0.15387



Perturbation Faithfulness (n=80 MRIs):
- Attention-guided deletion drops confidence faster than random (+0.12455 AUC).
- Attention-guided insertion restores confidence faster (+0.15387 AUC).
- 95% bootstrap CIs (2,000 resamples) strictly > 0 (p < 0.05).

FIG C: CLINICAL ViT ATTENTION VS CNN GRAD-CAM
Superior Medial Temporal Focus (ADNI Test)



Anatomical Localization Advantage:
- Baseline ViT scatters focus onto peripheral skull/scalp borders.
- CNN Grad-CAM activates on coarse local edges (64.20% acc).
- Fine-Tuned ViT-B/32 isolates hippocampal & ventricular atrophy.

06-07 SECONDARY SUBSYSTEMS: REALTIME VAULT, DOCKER API & ABLATION

WebSocket Sync & Containerisation

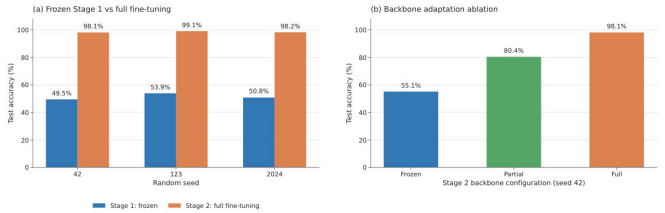
06. REAL-TIME TELEMETRY VAULT & EXPORT SUBSYSTEM

- Supabase Realtime Postgres Changes: client/src/app/results/page.tsx (L35-L44) subscribes to channel ('scans-db-changes') on public.scans (event: '*'), triggering zero-latency UI state hydration whenever a scan is analyzed or deleted across clinician devices.
- Clinical Reporting & CSV Batch Export: Implements client-side blob serialization (exportToCSV) exporting Date, File Name, Predicted Stage, Max Confidence, and all 4 class probabilities, alongside native print media queries and dynamic per-scan drill-down routes (/results/[id]).
- Longitudinal Analytics Dashboard: client/src/app/dashboard/page.tsx computes real-time aggregate KPIs (Total Neural Scans, Latest Diagnostic Verdict, Mean Confidence %, and Modal Stage) with instant search.

07. DUAL-CLOUD ASGI CONTAINERISATION & DEPENDENCY LOCK

- Pinned Deterministic Build (server/requirements.txt): Strictly pins fastapi==0.115.12, tensorflow==2.15.0, keras==2.15.0, tf-keras==2.15.0, numpy==1.26.4, and protobuf==4.25.3 to prevent pip resolver OOM kills.
- Hugging Face Spaces Docker Engine (alz-detect-api-hf/Dockerfile): Multi-layer python:3.11-slim image with libgl1-mesa-glx / libglib2.0-0 native OpenCV bindings, TF_USE_LEGACY_KERAS=1, Port 7860 ASGI loop, startup model warmup (np.zeros((1,224,224,3))), and /debug_v5 startup telemetry probe.

EMPIRICAL VISUAL PROOF: STAGE-1 VS STAGE-2 & BACKBONE ABLATION



Adaptation Regime / Seed	Trainable Params	Stage 1 Acc	Stage 2 Acc	Gain / Macro F1
3-Seed Mean (42, 123, 2024)	87.6M (Full ViT)	51.38+/-2.26%	98.47+/-0.53%	+47.10pp / 98.84%
Seed 42: Full 12-Block Tune	87.6M (Blocks 0-11)	49.45%	98.11% (Ep 30)	+48.66pp / 98.57%
Seed 42: Partial Blocks 8-11	28.5M (Top 4 Blks)	49.45%	80.44% (Ep 29)	+30.99pp / 83.69%
Seed 42: Frozen Backbone	197K (Head Only)	49.45%	55.11% (Ep 7)	+5.66pp / 47.37%

08 SECURITY MIDDLEWARE, DATA INTEGRITY & REPOSITORY TOPOLOGY

PostgreSQL RLS & Monospace Tree

MULTI-LAYER SECURITY & INTEGRITY MATRIX

1. PostgreSQL Row-Level Security (RLS)
`client/supabase/schema.sql`
Enforces ALTER TABLE public.scans ENABLE ROW LEVEL SECURITY with three cryptographic policies: Users can view (SELECT), insert (INSERT WITH CHECK), and delete (DELETE) strictly where auth.uid() = user_id. Prevents cross-tenant clinical data leakage.

2. Stateless Compute & Ephemeral RAM Purge
`server/src/routes/model_route.py (L525-L552)`
Executes inference inside explicit tf.device('/CPU:0') context with batch_size=1, encodes attention overlay to in-memory io.BytesIO Base64, closes all Matplotlib figures, and invokes gc.collect() to purge patient voxel tensors immediately.

3. SHA-256 Cryptographic Partition & Weight Audit
`Manuscript Sec. 2.2 & 2.7`
Conducted SHA-256 byte-level audit across all 40,384 development/test files (verifying zero byte-identical leakage between dev and 6,400-image test partitions) and verified seed-42 ViT checkpoint hashes prior to faithfulness testing.

4. CORS & Environment Secret Segregation
`server/src/server.py & docs/openai_api_fix_log.md`
Server-side OPENAI_API_KEY is isolated strictly inside Next.js server routes (/api/suggestions, /api/treatment) and never exposed to browser bundles; FastAPI enforces structured CORS and strict Pydantic/HTTPException error schemas.

PRODUCTION REPOSITORY FILE TREE (ANNOTATED MONOSPACE)

```
alz-detect/                          # Root Monorepo (Next.js + FastAPI + Weights)
|-- client/                          # Next.js 15.1 / React 19 / TypeScript Frontend
|   |-- src/app/                     # App Router Pages & Edge API Endpoints
|   |   |-- page.tsx                 # Landing Portal, Metrics & Live Demo Preview
|   |   |-- detect/page.tsx          # Diagnostic Upload, ValidationModal & XAI View
|   |   |-- dashboard/page.tsx       # Clinician KPI Telemetry & Temporal Log
|   |   |-- results/page.tsx         # Realtime Supabase Vault, Filter & CSV Export
|   |   |-- results/[id]/page.tsx    # Dynamic Per-Scan Diagnostic Drill-Down
|   |   |-- research/page.tsx        # ViT-B/32 Methodology & Clinical Protocol Docs
|   |   |-- ethics/ & privacy/       # Clinical AI Ethics & HIPAA/RLS Governance
|   |   `-- api/                     # Server-Side OpenAI Clinical Synthesis Routes
|   |       |-- suggestions/route.ts # POST /api/suggestions (Neuroradiology XAI)
|   |       `-- treatment/route.ts   # POST /api/treatment (Stage Care Protocol)
|   |-- src/contexts/AuthContext.tsx # Supabase Auth Session & Route Guard Provider
|   |-- src/lib/storage.ts           # Typed CRUD Wrapper for public.scans Table
|   `-- supabase/schema.sql          # DDL for public.scans, RLS Policies & Buckets
|-- server/                          # Primary FastAPI Inference Compute Cluster
|   |-- src/server.py                # ASGI Entry, KerasOpsShim, /debug_v5 & Warmup
|   |-- src/routes/model_route.py    # SafeViT-B/32, MRI Gate & Attention Rollout
|   |-- src/routes/root_route.py     # Healthcheck & Service Discovery Endpoint
|   `-- requirements.txt             # Pinned TF 2.15 / FastAPI 0.115 Lockfile
|-- alz-detect-api-hf/               # Hugging Face Spaces Docker Mirror (Port 7860)
|   `-- Dockerfile                   # python:3.11-slim + OpenCV GLX + Uvicorn CMD
`-- alz-detect-weights/              # Git-LFS Neural Checkpoint Store
    `-- RimanBassey_model.h5         # Fine-Tuned ViT-B/32 Weights (98.84% Test Acc)
```

09 CLOUD DEPLOYMENT ECOSYSTEM, VERIFICATION URLS & RESEARCH IMPACT

Multi-Cloud Production & Peer Review

CLOUD DEPLOYMENT & VERIFICATION MATRIX

Primary Platform (Production Domain)
`https://www.alzdetect.live/`
Custom clinical domain with global SSL/TLS, edge caching, and automated Git CI/CD deployments.

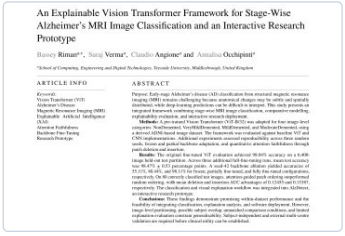
Diagnostic Scanner & Research Portal (Edge)
`https://www.alzdetect.live/detect | /research`
Next.js 15 App Router serverless runtime hosting dynamic OpenAI clinical inference routes.

Source Code & Weights Repository (GitHub)
`https://github.com/basseyriman/AlzDetect`
Full monorepo containing Next.js client, FastAPI server, Docker HF mirror, and Git-LFS weights.

ASGI Inference Cluster (Render / HF Spaces)
`POST /model/predict | GET /debug_v5`
Containerised Python 3.11/3.12 Uvicorn cluster pre-loading RimanBassey_model.h5 on startup.

Database & Auth Vault (Supabase Cloud)
`https://ebtnmkreczovteiyhgkd.supabase.co`
Managed PostgreSQL 15 with Row-Level Security, JWT authentication, and Realtime WebSockets.

SCIENTIFIC DISSEMINATION & JOURNAL SUBMISSION (ELSEVIER)


An Explainable Vision Transformer Framework for Stage-Wise Alzheimer's MRI Image Classification and an Interactive Research Prototype

Bassey Riman*, Suraj Verma, Claudio Angione and Annalisa Occhipinti

ARTICLE INFO

ABSTRACT

Title: An Explainable Vision Transformer Framework for Stage-Wise Alzheimer's MRI Image Classification and an Interactive Research Prototype
Lead Architect & First Author: Bassey Riman* (Corresponding Author), in collaboration with co-authors Suraj Verma, Claudio Angione, and Annalisa Occhipinti of Teesside University, UK
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Live Platform: https://www.alzdetect.live/



EDITORIAL MANAGER PORTAL & REPRODUCIBILITY VERIFICATION

Verified Author Submission Record (editorialmanager.com/neuri): Initial submission Oct 11, 2026; Revised manuscript NEURI-D-26-00238R1 logged Oct 16, 2026 ('Revisions Being Processed for Author' / Under Review). Backed by 3-seed reproducibility runs (Seeds 42, 123, 2024) and SHA-256 audit of 40,384 MRI slices.

Submission Portal: editorialmanager.com/neuri (Elsevier)
Primary URL: https://www.alzdetect.live/
Manuscript Ref: NEURI-D-26-00238R1 (20pp, 8 Figs)
Corpus Audit: 40,384 SHA-256 Verified MRIs